Part C

The most notable issue is the lack of random seed. Without a random seed NumPy will randomly choose any seed that will result in different outcomes; even with the same computer, everytime that the code it runs it will output something different. Hardcoding the absolute path file will make it difficult for the code to be run on the colleague's computer since they may have a different computer that used a different directory system (eg. Mac and Windows). The file would only be saved if the colleague has the same computer and exact same names for the directory. Moreover, the function that is used (np.random.randn) is part of NumPy's legacy random generation frameworks, but outcomes are more reproducible if they used np.random.default_rng() instead.

```
import numpy as np

# 1. Initialize a modern random number generator with a fixed seed
rng = np.random.default_rng(seed = 1)

# 2. Simulate returns (100 rows, 3 columns) using the seeded generator
data = rng.standard_normal((100, 3))

# 3. Take 20 random rows using the same seeded generator
# We use replace=True to mirror the original np.random.choice behavior
sample_indices = rng.choice(100, size = 20, replace = True)
sample = data[sample_indices]

# 4. Calculate and print the mean
print("mean of sample:", sample.mean())

# Note: If saving to a file, use relative paths or os.path.expanduser('~')
# to ensure it works seamlessly across different users and operating systems.
```

Part D

Reproducibility is incredibly important in financial analysis because they are often working with numbers that can greatly affect the standing of the business and drive critical decisions. If an auditor or someone who is looking at the numbers, they may not understand the context of where the final outputs are derived from in an email file, but a git commit shows the context by having line-by-line modifications. Being able to see the changes the file has undergone creates an auditable trail that allows for reproducibility and verification of the logic used to create the outputs.